

branch lengths
- see Pearse & paper

$$C = \begin{bmatrix} b_1 & b_{1,2} \\ b_{2,1} & b_2 \\ & & \ddots \end{bmatrix}$$

$$C_{BM} = \begin{pmatrix} \sigma_{BM}^2 \end{pmatrix} \begin{bmatrix} b_1 & & \\ & b_2 & \\ & & \ddots \end{bmatrix} = \sigma^2 \cdot C$$

$$= \begin{bmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_2^2 \\ & & \ddots \end{bmatrix} = \begin{cases} \text{Weir notation} \\ \text{phylocomm. uses} \\ \text{where } \sigma^2 \text{ is diff. than } \sigma \end{cases}$$

σ_{BM}^2 is the BM rate, all other are (co)vars

So... Morales-Castilla et al. 2022 notation is wrong given how we explain it but could be correct if all the σ s were squared

these off-diag. need to be σ^2

$$\begin{bmatrix} \sigma^2 & \sigma_{1,2}^2 \\ \sigma_{2,1}^2 & \sigma^2 \\ & & \ddots \end{bmatrix}$$

σ^2 is the rate of trt. ev!

$$\sigma = \sqrt{\frac{\text{trt}}{\text{time}}}$$

σ^2 :

future Will:
this covar mat $\times \sigma^2$
 $\Rightarrow$ all fns need σ^2